

Assignment: From Design to Implementation Java GUI Project with LLM Assistance

Course: Software Design & Analysis

Previous Work: Use case modeling, requirements specification, class diagram and similar

New Phase: Implementation in Java with GUI

Tools Allowed: Any LLM + IDE of choice

Objective

Implement the software system you designed earlier as a working Java application with a graphical user interface (GUI). You may use any LLM to accelerate coding, but you must critically analyze your experience with the LLM.

Part 1 – Implementation Requirements

1. **Java language only** – no other JVM languages (Kotlin, Scala, etc.).
2. **Graphical User Interface (GUI)**
 - Use Swing, JavaFX, AWT or any thing...
 - GUI must reflect at least two to three core use cases from your previous assignment.
3. **Mapping from design to code**
 - Your design must be clearly visible in your implementation (classes, attributes, methods, relationships like association, inheritance).
4. **Functionality**
 - Implement all functional requirements identified earlier.
 - Handle basic errors (e.g., invalid input, missing data) with user-friendly messages.
5. **Submission**
 - Complete project in github , submit link , keep it public.
 - Make video of project demo.

Part 2 – Technical Reflection

Write a short video (3 minutes) addressing the following four specific questions.

2.1 Why did you choose this particular LLM to increase productivity of writing code?

Explain your choice with technical reasons – examples:

- Better Java/Swing code generation in testing.
- Faster context understanding of class diagrams.
- Ability to handle GUI layout code (e.g., GridBagLayout, FXML).
- Integration with your IDE (e.g., Copilot in VS Code/IntelliJ).
- Cost, speed, or offline availability.

2.2 What was good and bad in this LLM-assisted approach?

SDA Assignment

Engr. Muhammad Umer Haroon

Add at least **3 good** and **3 bad** observations.

2.3 Ask your LLM: “Recommend the best design patterns for my project”

- Tell why those patterns were suggested and did you agreed or not ?

2.4 Interrogate and understand why these patterns

For each of the patterns recommended, answer:

1. What specific problem in your project does this pattern solve? (Connect to your use cases/requirements.)
2. Would your code work without this pattern? (Yes/no, and why.)
3. One limitation of using that pattern in your specific system.

Submission Checklist

- Java source code (working GUI). README.md (compile & run instructions) in git with public link.
- Video of project.
- Your video (for group of three combined video of around 3 minutes)

Academic Note

Using LLMs is allowed, but you must understand code submitted. If asked, you must explain your code and your LLM’s suggestions verbally. Copy-pasting without understanding is a violation of academic integrity.